
Healthcare Data Science Portfolio

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Former Pharmacist | Data Scientist | System Engineer

R • Python • SQL • Statistics • Healthcare Analytics

Combining clinical knowledge with data science to improve healthcare through reproducible analytics and evidence-based decision making.

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About Me

Pharmacist

Developed a strong foundation in clinical decision-making, evidence-based medicine and patient safety. Working closely with patients and healthcare professionals gave me an appreciation for the importance of accurate data and thoughtful interpretation in improving healthcare outcomes.

System Engineer

Strengthened my analytical problem-solving skills while working with large-scale IT systems. Gained experience in troubleshooting complex issues, maintaining reliable systems, documenting technical processes, and communicating effectively with diverse stakeholders.

Data Scientist

Applied statistics and machine learning to healthcare datasets through independent projects involving pharmacovigilance, clinical trial analysis, forecasting, and cancer genomics. Developed a strong interest in reproducible research and evidence-driven healthcare analytics.

My background combines healthcare, software engineering and data science. After working as a pharmacist, I became increasingly interested in how data can improve clinical decision-making and patient outcomes. This led me to pursue data science, where I developed skills in statistics, machine learning and reproducible research using R.

I enjoy working with healthcare data because it allows me to combine analytical thinking with clinical knowledge. Through independent projects involving pharmacovigilance, clinical trial analysis and cancer genomics, I have continued to develop practical experience applying statistical methods and machine learning to real healthcare questions.

I am particularly interested in healthcare analytics, real-world evidence, pharmacometrics and AI applications that improve patient care.

Technical Skills

Programming & Data Analysis

- R / tidyverse / ggplot2
- Python
- SQL
- Quarto / R Markdown
- Git / GitHub

Statistics & Machine Learning

- Data cleaning and wrangling
- Statistical analysis
- Regression and classification
- Time series forecasting
- Machine learning workflows

Healthcare & Research Interests

- Pharmacovigilance
- Drug safety
- Real-world data
- Pharmacometrics
- Healthcare analytics
- Medical AI

My goal is to combine healthcare knowledge, statistics, and machine learning to develop data-driven solutions that improve patient care.

Healthcare Analytics

Exploring Potential Drug Safety Signals using the FDA Adverse Event Reporting System (FAERS)

Full Report https://reinasdatastudio.github.io/faers-pharmacovigilance-analysis/FAERS_Report.html

GitHub Repository <https://github.com/reinasdatastudio/faers-pharmacovigilance-analysis>

Item	Details
Objective	Can publicly available FDA adverse event data identify potential safety signals?
Dataset	FDA FAERS
Tools	R, tidyverse, ggplot2, Quarto
Methods	Data cleaning, joins, ROR, visualisation
Outcome	Developed an end-to-end pharmacovigilance workflow for large-scale adverse event analysis.
Key Skills	Pharmacovigilance, reproducible analysis, statistical modelling

Objective

Can publicly available FDA adverse event data be used to identify potential drug safety signals?

This project investigated whether adverse drug reaction signals could be detected using the FDA Adverse Event Reporting System (FAERS), a large real-world pharmacovigilance database containing millions of spontaneous adverse event reports.

Specific objectives included:

- Cleaning and integrating multiple FAERS dataset using R
- Identifying commonly reported adverse events
- Exploring associations between selected adverse events and medications
- Calculating Reporting Odds Ratio (RORs) for selected drug-event combinations
- Demonstrating a reproducible pharmacovigilance workflow using publicly available healthcare data

Dataset

Source

FDA Adverse Event Reporting System (FAERS): <https://www.fda.gov/drugs/fda-adverse-event-monitoring-system-aems/fda-adverse-event-monitoring-system-aems-latest-quarterly-data-files>

Files Used

- DEMO25Q3.txt
- DRUG25Q3.txt
- REAC25Q3.txt

Methods

The analysis was conducted entirely in **R** using reproducible workflows.

Major steps included:

- Imported and cleaned multiple quarterly FAERS datasets
- Processed large text-based datasets and handled duplicate case reports
- Joined demographic, drug and reaction tables using patient identifiers
- Performed exploratory analyses to investigate adverse event frequencies
- Calculated Reporting Odds Ratios (ROR) to identify potential safety signals
- Produced publication-quality visualisations using **ggplot2**
- Generated a fully reproducible report using **Quarto**

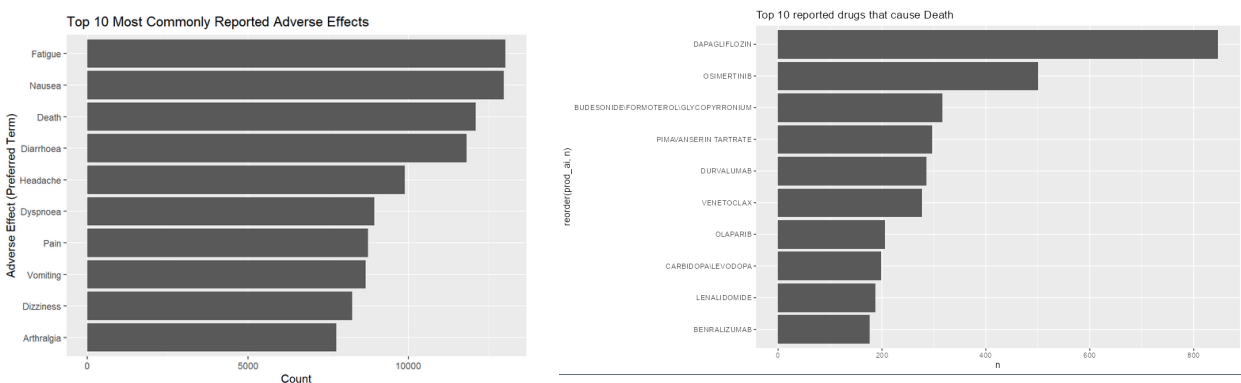
Results

The project successfully demonstrated an end-to-end pharmacovigilance workflow, from raw data processing through statistical signal detection.

Example outputs included:

- Most frequently reported adverse events
- Drug-specific adverse event summaries
- Reporting Odds Ratio (ROR) calculations
- Safety signal visualisations

The project also explored the strengths and limitations of spontaneous reporting systems, including reporting bias, duplicate reports, and the inability to establish causality.



Skills Demonstrated

Data Science

- Large-scale data cleaning
- Data integration
- Exploratory data analysis
- Statistical analysis
- Data visualisation

Healthcare

- Pharmacovigilance
- Drug safety
- FDA FAERS database
- Healthcare data interpretation

Technical

- R
- tidyverse
- ggplot2
- Quarto
- Reproducible reporting

What I Learned

This project strengthened both my technical and healthcare analytics skills by applying statistical methods to a real-world healthcare dataset. It reinforced the importance of reproducible workflows, careful data cleaning, and critical interpretation of observational healthcare data.

It also deepened my interest in pharmacovigilance, real-world data, and the application of data science to improve medication safety and healthcare decision-making.

Reproducible Pharmacovigilance Workflow Using FAERS

Full Report <https://reinasdatastudio.github.io/faers-reproducible-workflow/FAERS2025Q3.html>

GitHub <https://reinasdatastudio.github.io/faers-reproducible-workflow>

Item	Details
Objective	Investigate adverse event reporting pattern while demonstrating best practices for reproducible healthcare data analysis
Dataset	FDA FAERS
Tools	R, tidyverse, ggplot2, Quarto
Methods	Data cleaning, joins, visualisation
Outcome	Developed an end-to-end pharmacovigilance workflow for large-scale adverse event analysis.
Key Skills	Pharmacovigilance, reproducible analysis, statistical modelling

Project Overview

This project demonstrates an end-to-end pharmacovigilance workflow using the FDA Adverse Event Reporting System (FAERS) 2025 Q3 dataset.

The objective was to investigate adverse event reporting patterns while demonstrating best practices for reproducible healthcare data analysis. Rather than estimating clinical risk, the analysis focused on exploratory signal awareness through careful case-level aggregation, transparent data processing, and cautious interpretation of spontaneous reporting data.

Dataset

Source

FDA Adverse Event Reporting System (FAERS): <https://www.fda.gov/drugs/fda-adverse-event-monitoring-system-aems/fda-adverse-event-monitoring-system-aems-latest-quarterly-data-files>

Data Used

- Reaction (REAC25Q3.txt)
- Outcome (OUTC25Q3.txt)

Approximately **1.5 million adverse reaction records** from the 2025 Q3 public release were processed and analysed.

Analytical Workflow

The analysis followed a fully reproducible workflow implemented in **R**.

Data Preparation

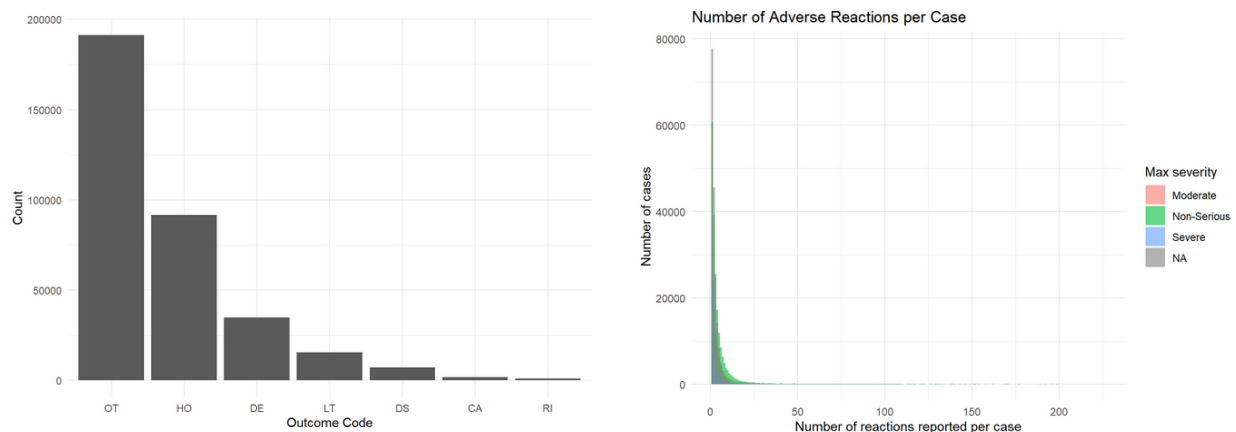
- Imported raw FAERS ASCII datasets
- Cleaned and transformed multiple healthcare datasets
- Performed case-level aggregation to minimise duplicate follow-up reports
- Joined reaction and outcome datasets using unique identifiers

Exploratory Analysis

- Investigated adverse reaction reporting behaviour
- Examined the distribution of reported reaction terms
- Analysed outcome severity across reported cases
- Combined reaction frequency with outcome severity to provide additional clinical context

Reporting

- Generated publication-quality tables and figures
- Produced a full Quarto report directly from the analysis code



Key Findings

The analysis demonstrated several important characteristics of spontaneous adverse event reporting systems:

- Most cases contained only a small number of reported adverse reactions.
- Frequently reported reaction terms were not necessarily associated with the most severe outcomes.
- Cases reporting larger numbers of adverse reactions showed a slightly higher proportion of severe outcomes.
- Outcome information was incomplete for many reports, highlighting the importance of cautious interpretation when working with real-world pharmacovigilance data.

Skills Demonstrated

Healthcare Analytics

- Pharmacovigilance
- Drug safety analysis
- Healthcare data interpretation
- Clinical data reporting

Data Science

- Large-scale data cleaning
- Data integration
- Exploratory data analysis
- Statistical summarisation
- Data visualisation

Technical

- R
- tidyverse
- ggplot2
- Quarto
- Reproducible research workflows

What I Learned

This project was one of my first project using FAERS. It strengthened my ability to work with large healthcare datasets while maintaining reproducible analytical practices. It reinforced the importance of understanding the limitations of observational healthcare data, documenting every processing step, and communicating results responsibly. The project also deepened my interest in pharmacovigilance, real-world evidence, and healthcare data science.

Clinical Trial Analysis of Hypertension Treatments

Full Report <https://reinasdatastudio.github.io/clinical-trial-hypertension-analysis/clinical-trial-hypertension-analysis.html>

GitHub <https://reinasdatastudio.github.io/clinical-trial-hypertension-analysis>

Item	Details
Objective	Compare the effectiveness of two antihypertensive treatments using statistical analysis.
Dataset	Synthetic Phase III Clinical Trial Dataset (1,000 patients)
Tools	R, tidyverse, ggplot2, Quarto
Methods	Data cleaning, EDA, Logistic Regression, Non-parametric Tests
Outcome	Evaluated treatment efficacy and identified factors associated with persistent hypertension
Key Skills	Clinical Statistics, Healthcare Analytics, Data Visualisation, Reproducible Research

Project Overview

This project analysed a synthetic Phase III clinical trial dataset to evaluate the effectiveness of antihypertensive treatments using statistical methods commonly applied in clinical research.

The objective was to compare treatment outcomes between two investigational drugs and a placebo while demonstrating a reproducible workflow for clinical data cleaning, exploratory analysis, statistical testing, and regression modelling in R.

Although the dataset was synthetic, the analytical workflow was designed to reflect the approach used when analysing real clinical trial data.

Dataset

Source

Synthetic Phase III Clinical Trial Dataset: <https://www.kaggle.com/datasets/isabelladil/phase-iii-clinical-trial-dataset/data>

Study Population

- 1,000 patients
- Multiple clinical study sites
- Three treatment groups (Drug A, Drug B, Placebo)

Variables Included

- Demographics (Age, Gender)
- Systolic and Diastolic Blood Pressure
- Cholesterol Level
- Adverse Events
- Treatment Allocation
- Patient Dropout Status

Analytical Workflow

The analysis was conducted using a fully reproducible workflow in **R**.

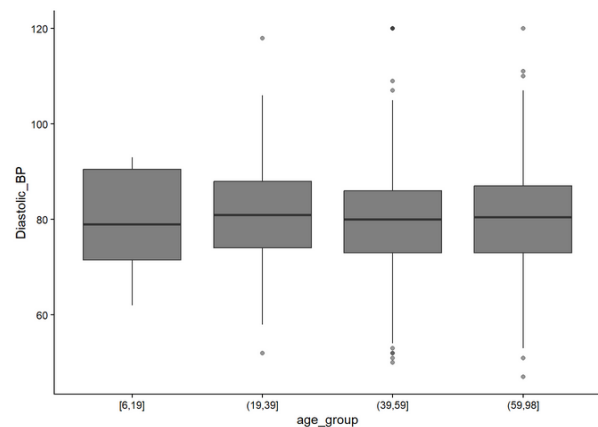
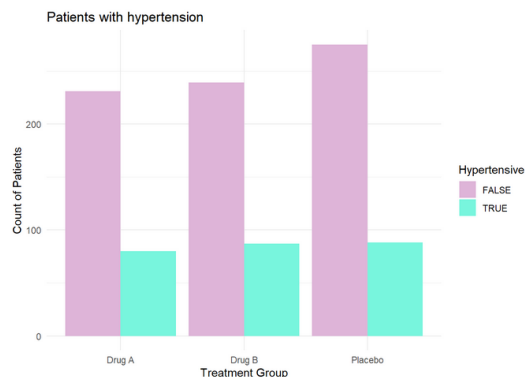
Data Preparation

- Assessed data quality and completeness
- Confirmed no missing values
- Identified cholesterol outliers using age-stratified IQR methods
- Created clinically meaningful age groups
- Defined hypertension using blood pressure thresholds consistent with clinical practice

Exploratory Data Analysis

- Examined treatment group balance
- Visualised blood pressure and cholesterol distributions
- Explored relationships between age, cholesterol, and blood pressure
- Investigated treatment response and patient dropout patterns

```
Treatment_Group mean_sys sd_sys mean_dia n
<chr>          <dbl>   <dbl>   <dbl> <int>
1 Drug A         120.    15.4    80.3  311
2 Drug B         119.    14.4    80.3  326
3 Placebo        119.    17.4    80.0  363
```



Statistical Analysis

- Kruskal–Wallis tests for comparing treatment groups
- Pairwise Wilcoxon tests with Benjamini–Hochberg correction
- Chi-squared test for hypertension prevalence
- Multivariable logistic regression to identify predictors of persistent hypertension
- Multicollinearity assessment using Variance Inflation Factors (VIF)

Key Findings

The statistical analyses consistently indicated no significant differences in blood pressure outcomes between the two active treatments and the placebo group.

Additional findings included:

- Treatment groups were well balanced at baseline.
- None of the evaluated patient characteristics significantly predicted persistent hypertension after treatment.
- Cholesterol level showed only a weak, non-significant association with persistent hypertension.
- The project highlighted the importance of objective statistical evaluation rather than relying on apparent differences observed during exploratory analysis.

Skills Demonstrated

Clinical Data Analysis

- Clinical trial data interpretation
- Treatment efficacy evaluation
- Healthcare data visualisation
- Evidence-based statistical interpretation

Statistics

- Exploratory Data Analysis (EDA)
- Logistic regression
- Kruskal–Wallis test
- Wilcoxon rank-sum test
- Chi-squared test
- Multicollinearity assessment

Technical

- R
- tidyverse
- ggplot2
- Quarto
- Reproducible research workflows

What I Learned

This project strengthened my understanding of clinical trial analysis and the importance of selecting statistical methods appropriate for healthcare data. It reinforced the value of reproducible workflows, transparent reporting, and interpreting negative findings objectively.

The project also improved my ability to translate statistical results into clinically meaningful conclusions, an essential skill for healthcare research and data science.

Machine Learning for Cancer Variant Pathogenicity Classification

Kaggle Notebook: <https://www.kaggle.com/code/reinagraham/cancer-variant-dataset-machine-learning-using-r>

Quarto html <https://reinasdatastudio.github.io/cancer-variant-machine-learning/cancer-variant-dataset-machine-learning.html>

Item	Details
Objective	Classify genetic cancer variants according to their clinical significance using supervised machine learning.
Dataset	Cancer Variant Prioritization Dataset (10,000 genetic variants)
Tools	R, tidyverse, randomForest, ggplot2, Quarto
Methods	Data Cleaning, Exploratory Data Analysis, Logistic Regression, Random Forest, Class Imbalance Assessment, Model Evaluation
Outcome	Compared traditional statistical modelling and machine learning approaches to classify pathogenic variants and evaluated their performance on an imbalanced healthcare dataset.
Key Skills	Machine Learning, Healthcare Analytics, Genomic Data Analysis, Model Evaluation, Reproducible Research

Project Overview

This project investigated the use of supervised machine learning to classify genetic variants according to their clinical significance using a publicly available cancer genomics dataset.

The objective was to compare traditional statistical modelling with machine learning approaches while developing a reproducible workflow for data preprocessing, exploratory analysis, model development, and evaluation. Particular emphasis was placed on understanding the challenges of class imbalance and model interpretation in a healthcare setting.

Dataset

Source

Cancer Variant Prioritization Dataset (Kaggle): <https://www.kaggle.com/code/reinagraham/cancer-variant-dataset-machine-learning-using-r/input>

Dataset Characteristics

- 10,000 genetic variants
- Clinical significance classified as **Benign**, **Pathogenic**, or **Variant of Uncertain Significance (VUS)**
- Genomic and predictive annotation features including:
 - Gene
 - Variant type
 - Population allele frequency

- CADD score
- SIFT prediction
- PolyPhen prediction

Analytical Workflow

The analysis was performed using a fully reproducible workflow in **R**.

Data Preparation

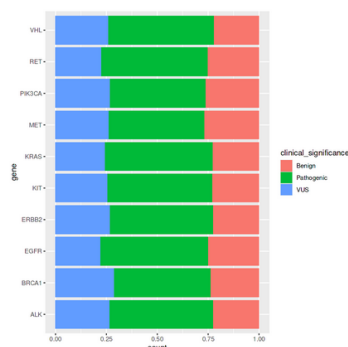
- Inspected data quality and verified completeness
- Cleaned variable names and converted variables to appropriate data types
- Combined *Likely Pathogenic* and *Pathogenic* categories to reduce label ambiguity
- Removed duplicate observations
- Evaluated potential outliers while considering biological relevance rather than removing values solely based on statistical thresholds

Exploratory Data Analysis

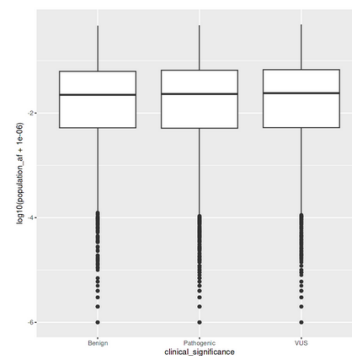
- Investigated distributions of genomic features
- Explored relationships between allele frequency, CADD score, variant type, and clinical significance
- Demonstrated that individual variables showed substantial overlap between clinical significance groups, motivating the need for multivariable predictive models

```
In [19]: top_genes <- df %>%
  count(gene, sort = TRUE) %>%
  slice_head(n = 10) %>%
  pull(gene)

df %>%
  filter(gene %in% top_genes) %>%
  ggplot(aes(x = gene, fill = clinical_significance)) +
  geom_bar(position = "fill") +
  coord_flip()
```



```
In [20]: df %>%
  ggplot(aes(x = clinical_significance, y = log10(population_af + 1e-6))) +
  geom_boxplot()
```



Model Development

Two classification approaches were evaluated:

- Logistic Regression (Generalised Linear Model)
- Random Forest

Model performance was assessed using both overall accuracy and class-specific prediction performance to better understand the impact of class imbalance.

Key Findings

Several important observations emerged during the analysis:

- Individual genomic features alone provided limited discriminatory power for distinguishing pathogenic variants.
- Logistic regression achieved reasonable overall accuracy but classified nearly all observations as **Pathogenic**, demonstrating that accuracy alone can be misleading for imbalanced healthcare datasets.
- Random Forest better captured non-linear relationships and improved class balance, although bias towards the majority class remained.
- Introducing class weighting highlighted the trade-off between overall accuracy and fair classification across classes.
- The analysis emphasised the importance of evaluating precision, recall, and class-specific performance rather than relying solely on overall accuracy.

Skills Demonstrated

Machine Learning

- Binary classification
- Logistic Regression (GLM)
- Random Forest
- Hyperparameter tuning
- Feature importance analysis
- Model comparison

Data Science

- Data cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering
- Class imbalance assessment
- Model evaluation
- Statistical interpretation

Technical

- R
- tidyverse
- randomForest
- ggplot2
- Reproducible research workflows

What I Learned

This project strengthened my understanding of applying machine learning to healthcare data, particularly the challenges associated with imbalanced clinical datasets. Rather than focusing solely on predictive accuracy, I learned the importance of selecting evaluation metrics that reflect real-world performance and carefully interpreting model behaviour.

The project also reinforced the value of combining statistical reasoning with machine learning techniques to develop transparent, reproducible analytical workflows suitable for healthcare applications.

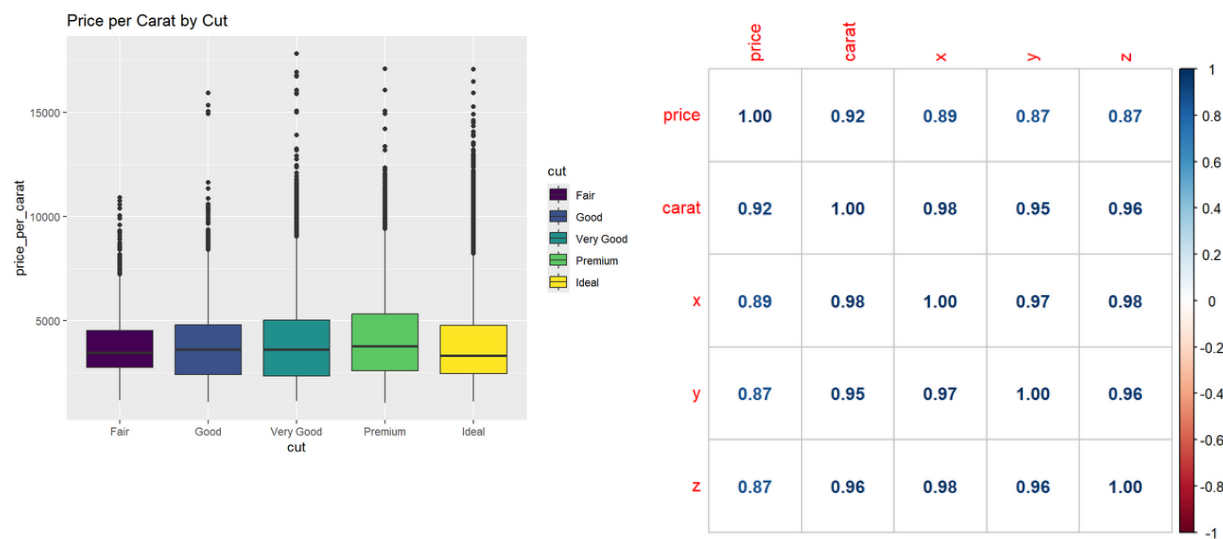
Predictive Analysis

Diamond Price Prediction using Statistical and Machine Learning Models

Full Report: <https://reinasdatastudio.github.io/diamond-price-prediction/diamonds-price-prediction.html>

GitHub Repository: <https://reinasdatastudio.github.io/diamond-price-prediction>

Item	Details
Objective	Develop predictive models to estimate diamond prices and compare the performance of traditional statistical modelling with machine learning.
Dataset	ggplot2 Diamonds Dataset ($\approx 54,000$ observations)
Tools	R, tidyverse, ggplot2, ranger, Quarto
Methods	Data Cleaning, Feature Engineering, Exploratory Data Analysis, Correlation Analysis, Multiple Linear Regression, Random Forest Regression, Model Diagnostics
Outcome	Developed and compared regression models, demonstrating that Random Forest provided improved predictive performance while linear regression offered greater interpretability.
Skills Demonstrated	Predictive Modelling, Regression Analysis, Machine Learning, Feature Engineering, Model Evaluation, Data Visualisation, Reproducible Research



Project Overview

This project investigated the factors that influence diamond prices using the well-known **ggplot2 Diamonds** dataset containing nearly **54,000 observations**. The objective was to develop predictive models capable of estimating diamond prices while comparing the strengths and limitations of traditional statistical modelling and machine learning approaches. The project followed a complete end-to-end workflow including data cleaning, feature engineering, exploratory data analysis, predictive modelling, and model evaluation.

Dataset

Source

ggplot2 Diamonds Dataset

Dataset Characteristics

- Approximately 54,000 diamonds
- Price range from \$326 to \$18,823
- Predictor variables including:
 - Carat
 - Cut
 - Colour
 - Clarity
 - Physical dimensions (x, y, z)
 - Depth
 - Table percentage

Analytical Workflow

Data Preparation

- Removed observations with impossible physical measurements (zero length, width, or depth)
- Verified data completeness and confirmed no missing values
- Created engineered features including **price per carat**
- Applied logarithmic transformations to reduce skewness and improve linear model assumptions
- Retained legitimate high-value diamonds rather than removing statistical outliers to avoid biasing predictions for premium diamonds

Exploratory Data Analysis

Exploratory analysis examined both numerical and categorical variables to understand the factors associated with diamond prices.

Key analyses included:

- Distribution of price and carat
- Distribution of cut, colour, and clarity grades
- Price per carat across cut quality

- Correlation analysis of numerical variables
- Investigation of multicollinearity among size-related measurements

The analysis identified **carat** as the strongest predictor of price, while also demonstrating strong correlations between carat and physical dimensions (x, y, z).

Model Development

Two predictive models were developed and compared:

- Multiple Linear Regression
- Random Forest Regression (ranger)

The linear regression model incorporated logarithmic transformations to satisfy model assumptions, while the Random Forest model was trained using the original variables. Highly correlated variables were intentionally excluded to minimise multicollinearity and reduce the risk of overfitting.

Key Findings

The project demonstrated several important modelling insights:

- **Carat** was by far the strongest predictor of diamond price.
- Cut, colour, and clarity further improved predictive performance.
- Linear regression achieved excellent explanatory power (**Adjusted $R^2 \approx 0.983$**) but residual diagnostics revealed heteroscedasticity, indicating reduced reliability for high-priced diamonds.
- Random Forest produced superior predictive performance with a lower RMSE and a higher test-set $R^2 (\approx 0.977)$, particularly for expensive diamonds where non-linear relationships were more apparent.
- The project highlighted the importance of evaluating model assumptions in addition to predictive accuracy when selecting an appropriate modelling approach.

Skills Demonstrated

Machine Learning

- Regression modelling
- Random Forest Regression
- Model comparison
- Feature importance
- Model evaluation

Statistics

- Multiple Linear Regression
- Correlation analysis
- Residual diagnostics
- Assessment of heteroscedasticity
- Multicollinearity assessment

- Feature engineering

Technical

- R
- tidyverse
- ggplot2
- ranger
- Quarto
- Reproducible analytical workflows

What I Learned

This project strengthened my understanding of predictive modelling by demonstrating that selecting the most appropriate model requires more than simply maximising predictive accuracy. While linear regression provided an interpretable model with excellent overall performance, diagnostic evaluation revealed violations of model assumptions for high-value observations. Comparing these results with a Random Forest model reinforced the importance of balancing interpretability, statistical assumptions, and predictive performance when solving real-world analytical problems.

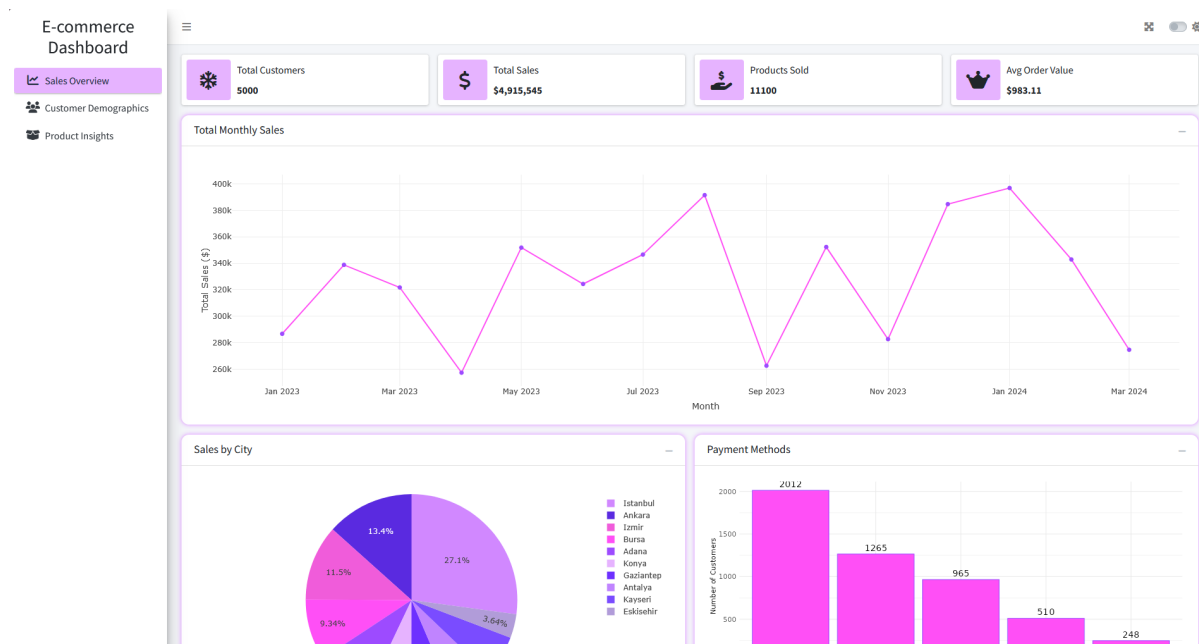
Business Intelligence and Data Visualisation with Interactive Dashboard

Interactive E-commerce Analytics Dashboard

Interactive Application: <https://thequeenofdata.shinyapps.io/ecommerceDashboardExample/>

GitHub Repository: <https://github.com/reinasdatastudio/ecommerce-sales-dashboard>

Item	Details
Objective	Design and develop an interactive business intelligence dashboard for exploring sales performance, customer behaviour, and product insights.
Dataset	E-commerce Customer Behaviour Dataset
Tools	R • Shiny • bs4Dash • Plotly • ggplot2 • DT • tidyverse
Methods	Interactive Dashboard Development • Data Cleaning • KPI Reporting • Reactive Programming • Data Visualisation
Outcome	Developed a multi-page dashboard enabling users to interactively explore business performance through dynamic filtering, real-time summaries, and interactive visualisations.
Skills Demonstrated	Dashboard Development • Business Intelligence • Reactive Programming • Data Visualisation • User Interface Design • Reproducible Research



Project Overview

This project involved the development of an interactive business intelligence dashboard using **R Shiny** and **bs4Dash** to analyse customer purchasing behaviour, sales performance, and product trends within an e-commerce dataset. The dashboard was designed to provide decision-makers with an intuitive interface for exploring key business metrics through interactive visualisations and real-time filtering.

Dashboard Features

The application consists of three integrated dashboard pages:

Sales Overview

Provides high-level business KPIs including:

- Total sales
- Total customers
- Products sold
- Average order value
- Monthly sales trends
- Sales distribution by city
- Payment method analysis

Customer Demographics

Explores customer behaviour through:

- Returning customer rate
- Customer ratings
- Monthly customer activity
- Gender and age distributions
- Geographic customer distribution
- Relationship between delivery time and customer ratings
- Interactive customer data table

Product Insights

Allows users to investigate product performance using:

- Interactive age-range filtering
- Top-selling product categories
- Best and lowest revenue-generating categories
- Monthly sales explorer
- Daily sales trends
- Top-performing products by sales and quantity
- Highest-selling products across cities

Technical Highlights

The dashboard demonstrates several important Shiny development concepts:

- Multi-page dashboard design using **bs4Dash**
- Reactive programming with dynamic filtering
- Interactive visualisations using **Plotly**
- Interactive data tables using **DT**
- Custom dashboard theming with **fresh**
- KPI cards and business reporting
- Responsive dashboard layout
- Efficient data preprocessing using **tidyverse**

Skills Demonstrated

Dashboard Development

- Shiny
- bs4Dash
- Reactive Programming
- User Interface Design

Data Visualisation

- ggplot2
- Plotly
- Interactive reporting

Data Analysis

- KPI reporting
- Customer segmentation
- Sales analytics
- Business intelligence

Technical

- R
- tidyverse
- DT
- fresh
- lubridate

What I Learned

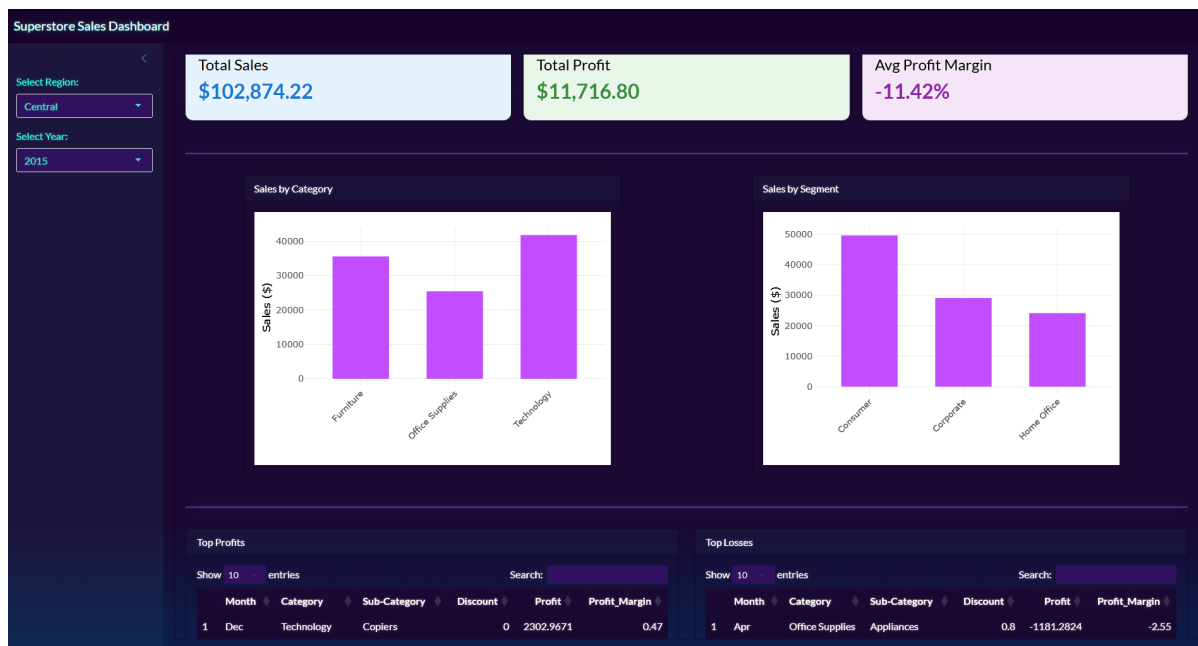
This project strengthened my ability to transform raw transactional data into an interactive dashboard that supports exploratory analysis and business decision-making. It also provided practical experience designing user-focused interfaces and implementing reactive workflows that allow users to investigate data dynamically rather than relying on static reports.

Interactive Superstore Business Intelligence Dashboard

Interactive Dashboard <https://thequeenofdata.shinyapps.io/Superstore/>

GitHub Repository <https://github.com/reinasdatastudio/superstore-business-intelligence-dashboard>

Items	Details
Objective	Develop an interactive business intelligence dashboard to analyse sales performance, profitability, and customer segments across different regions and years.
Dataset	Superstore Sales Dataset (cleaned transactional data)
Tools	R • Shiny • bslib • Plotly • ggplot2 • DT • dplyr
Methods	Interactive Dashboard Development • KPI Reporting • Reactive Programming • Data Aggregation • Business Intelligence
Outcome	Developed an interactive dashboard enabling users to explore sales trends, profitability, customer segments, and regional performance through dynamic filtering and interactive visualisations.
Skills Demonstrated	Dashboard Development • Business Intelligence • Reactive Programming • Data Visualisation • User Interface Design • Data Summarisation



Project Overview

This project involved the development of an interactive business intelligence dashboard using **R Shiny** to support the exploration of sales performance, profitability and customer segments within a retail dataset. The application enables users to dynamically filter results by **region** and **year**, providing an intuitive interface for analysing key business metrics and identifying trends that support data-driven decision-making.

Dashboard Features

The dashboard provides a range of interactive analytical tools, including:

Executive KPI Dashboard

- Total Sales
- Total Profit
- Average Profit Margin

Key performance indicators update automatically based on the selected region and year, allowing users to compare business performance across different markets.

Sales Performance Analysis

Interactive visualisations summarise sales across:

- Product Categories
- Customer Segments

Hover-enabled charts provide detailed values while allowing users to quickly compare performance between business areas.

Profitability Analysis

The dashboard includes searchable interactive tables highlighting:

- Highest-profit transactions
- Largest financial losses

This enables rapid identification of exceptional business performance and potential areas requiring further investigation.

Trend Analysis

Monthly profit trends are displayed using interactive time-series visualisations, allowing users to investigate seasonal patterns and changes in profitability throughout the selected year.

Technical Highlights

The application demonstrates several best practices in Shiny application development:

- Reactive filtering by region and year
- Pre-computed summary tables to improve application performance
- Interactive visualisations using Plotly
- Responsive dashboard interface using **bslib**
- Interactive data tables using **DT**
- Efficient data manipulation using **dplyr**
- Modular reactive workflow separating preprocessing, filtering, and visualisation logic

Skills Demonstrated

Dashboard Development

- R Shiny
- Reactive Programming
- User Interface Design
- Business Intelligence Dashboard Development

Data Analysis

- KPI Reporting
- Sales Analytics
- Profitability Analysis
- Customer Segmentation
- Trend Analysis

Data Visualisation

- ggplot2
- Plotly
- Interactive Reporting

Technical

- R
- dplyr
- lubridate
- bslib
- DT

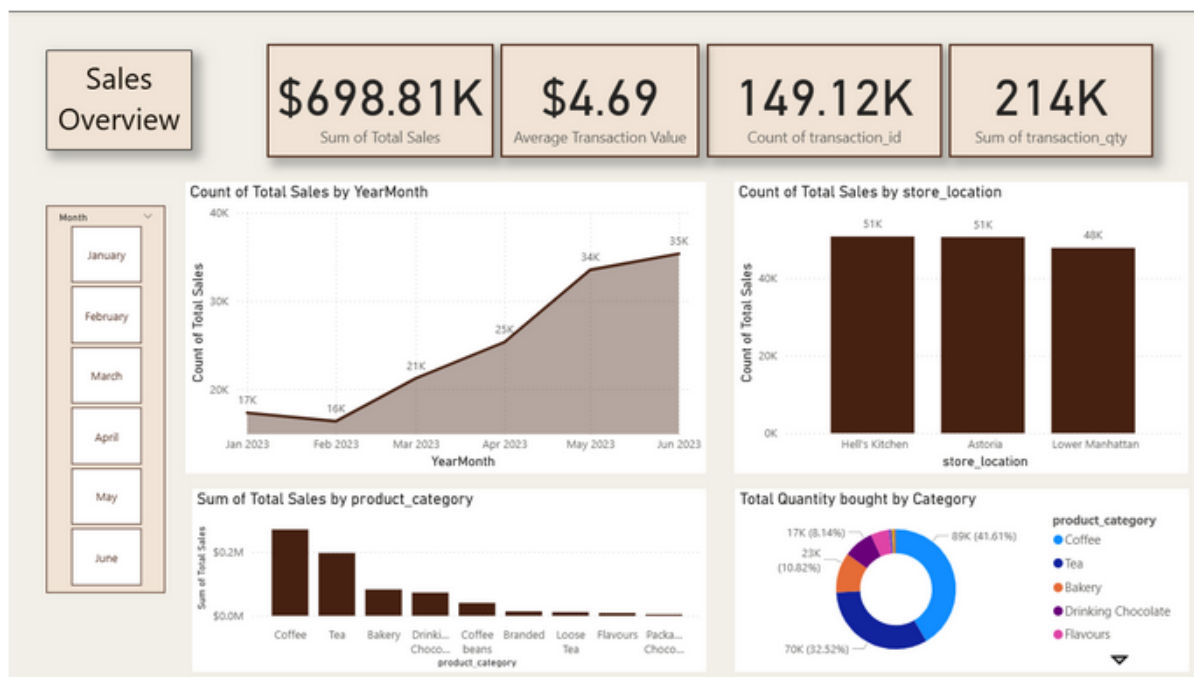
What I Learned

This project strengthened my understanding of interactive dashboard development by demonstrating how complex transactional data can be transformed into an intuitive decision-support tool. I gained practical experience designing responsive user interfaces, implementing efficient reactive workflows, and presenting business metrics through interactive visualisations that enable users to explore data dynamically rather than relying on static reports.

Coffee Shop Sales Analytics Dashboard (Power BI)

Full Report <https://reinasdatastudio.github.io/coffee-shop-powerbi-dashboard/coffee-shop-powerbi.html>

Item	Details
Objective	Design an interactive Power BI dashboard to analyse sales performance, product trends, and store operations for a multi-location coffee shop.
Dataset	Maven Analytics Coffee Shop Sales Dataset (149,117 transactions)
Tools	Power BI • Power Query • DAX
Methods	Data Cleaning • Data Transformation • KPI Reporting • Interactive Dashboard Design • Business Intelligence
Outcome	Developed a three-page interactive dashboard providing insights into sales trends, product performance, customer purchasing behaviour, and store-level operations.
Skills Demonstrated	Power BI • Business Intelligence • Dashboard Design • DAX • Data Modelling • Data Visualisation



Project Overview

This project involved the development of a three-page interactive business intelligence dashboard using **Power BI** to analyse sales performance for a multi-location coffee shop. The dashboard transforms transactional sales data into interactive visualisations that support business decision-making by highlighting revenue trends, product performance, customer purchasing behaviour, and store-level operational insights. The project incorporated data preparation using **Power Query**, analytical modelling with **DAX**, and dashboard design following business intelligence best practices.

Dashboard Features

The dashboard consists of three interactive reporting pages.

Sales Overview

Provides an executive summary of business performance including:

- Total sales and transaction volume
- Monthly sales trends
- Sales by store location
- Product category performance
- Product category contribution to overall sales
- Interactive month filtering

This page enables decision-makers to quickly assess overall business performance and identify high-level sales trends.

Product Performance

Focuses on detailed product analytics, including:

- Sales by product category
- Top-selling products
- Product quantity sold
- Sales by location and month
- Hourly purchasing behaviour
- Interactive category and month filtering

These visualisations help identify the most profitable product categories and reveal customer purchasing patterns throughout the day.

Store Performance

Examines operational performance across individual store locations through:

- Total sales by store
- Average transaction value
- Hourly sales heatmaps
- Sales trends by location
- Store-specific filtering

The dashboard supports comparison between store locations and highlights opportunities to optimise staffing and operational performance during different trading periods.

Technical Highlights

The project demonstrates practical experience with Power BI business intelligence workflows, including:

- Data cleaning and transformation using **Power Query**
- Time intelligence modelling using **DAX**
- Custom Date Table development
- KPI dashboard design
- Interactive slicers and filtering
- Conditional formatting
- Multi-page dashboard architecture
- Business-focused data storytelling

Skills Demonstrated

Business Intelligence

- Power BI Dashboard Development
- KPI Reporting
- Business Performance Analysis
- Sales Analytics

Data Preparation

- Power Query
- Data Cleaning
- Data Transformation
- Data Modelling

Analytics

- DAX
- Time Intelligence
- Customer Behaviour Analysis
- Product Performance Analysis

Visualisation

- Interactive Dashboards
- Executive Reporting
- Data Storytelling

What I Learned

This project strengthened my understanding of end-to-end business intelligence development using Power BI. I gained experience transforming raw transactional data into an interactive decision-support dashboard through data preparation, DAX modelling, and effective dashboard design. The project reinforced the importance of presenting complex analytical results through intuitive visualisations that enable stakeholders to explore business performance efficiently.

Reina's Data Studio

<https://reinasdatastudio.github.io/webpage/>

Overview

Reina's Data Studio is my independent data science and R consulting initiative focused on reproducible analysis, healthcare analytics, and data communication.

Activities

- Created a professional website using Quarto
- Published technical blog posts about R and reproducible analysis
- Developed healthcare data analysis projects
- Created educational content for R and data science
- Communicated technical concepts in both English and Japanese

Research Interests

I am particularly interested in applying data science to healthcare problems such as:

- Drug safety and pharmacovigilance
- Real-world healthcare data analysis
- Clinical decision support
- Pharmacometrics and PK/PD modelling
- Medical AI and responsible use of data
- Reproducible research workflows

Current Learning Goals

- Real-world data analysis using claims and electronic health records
- PK/PD modelling and pharmacometrics
- Python for healthcare data science
- Cloud-based data workflows
- AI/ML applications in healthcare

Contact

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